Magnetic Observations

Preserve raw evidence before interpreting a boundary

One sample, one timestamp, one stable snapshot

Lesson	034 — component	**Time**	120–180 minutes
Prerequisites	Lessons 002, 007, and 008	**Board**	Arduino Mega 2560
Evidence	TP-H, TP-R, three LED witnesses	**Status**	host verified; bench open
Energy class	E0 host; provisional E1 fixture after specimen qualification		

Specimen gate. The authorized inventory lists "Linear Hall" and "Magnetic Spring" families. It does not establish output type, pin order, voltage, pull-up rail, polarity, active level, transfer curve, or a safe magnetic stimulus. Keep both specimens unpowered and quarantined until their exact markings and primary electrical evidence resolve every role.

1 Mission and learning outcomes

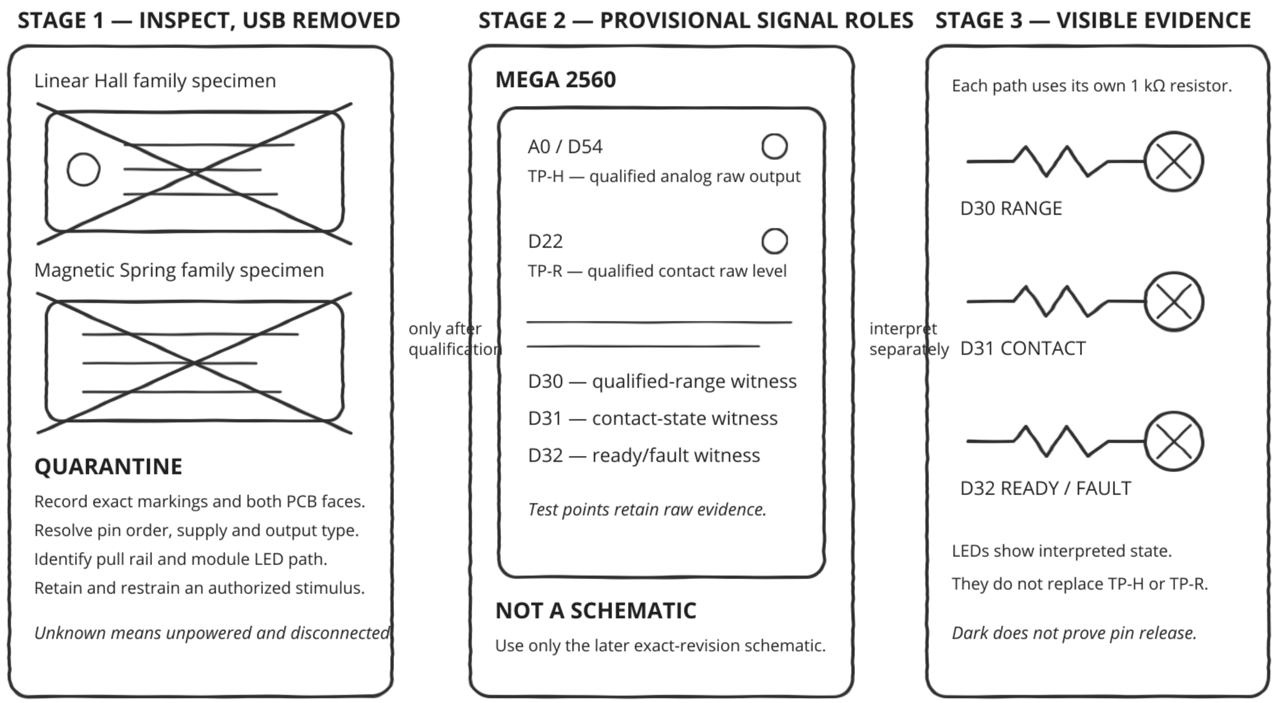
The lesson separates three layers:

$$\text{raw sample} \to \text{qualified range and dwell} \to \text{stable magnetic observation.}$$

By the end, you can:

- preserve an ADC count or contact level beside its interpretation;
- distinguish out-of-range evidence from an invented fault diagnosis;
- apply hysteresis and continuous dwell without blocking;
- explain why a digital contact has no inferred magnetic polarity;
- reject backward time while accepting natural timestamp rollover;
- prove resource acquisition separately from shutdown pin state.

2 Inspect before any powered circuit



Alternative text: pencil inspection plate showing two unidentified magnetic-family specimens crossed out and quarantined until qualified, then a Mega observation stage with TP-H and TP-R leading to three resistor-limited evidence LEDs.

The drawing is an orientation and decision path, not an electrical schematic. No electrically authoritative schematic is possible until the exact retained specimens are identified. With USB absent, record both PCB faces, markings, connector labels, continuity-safe ground evidence, supply rating, output topology, pull rail, module LED path, and maximum current. A cracked glass capsule, ambiguous mercury-like part, unresolved pinout, or loose magnetic stimulus stops the physical route.

Evidence point	Provisional Mega pin	Meaning after qualification
TP-H	A0 / D54	retained linear-family raw ADC output
TP-R	D22	retained contact-family raw digital output
range LED	D30 through 1 kΩ	TP-H lies inside configured range
contact LED	D31 through 1 kΩ	contact observation is active
ready/fault LED	D32 through 1 kΩ	acquisition and current health

3 Linear observation contract

`LinearHall` owns one `AnalogInput`. Its configured ordering is:

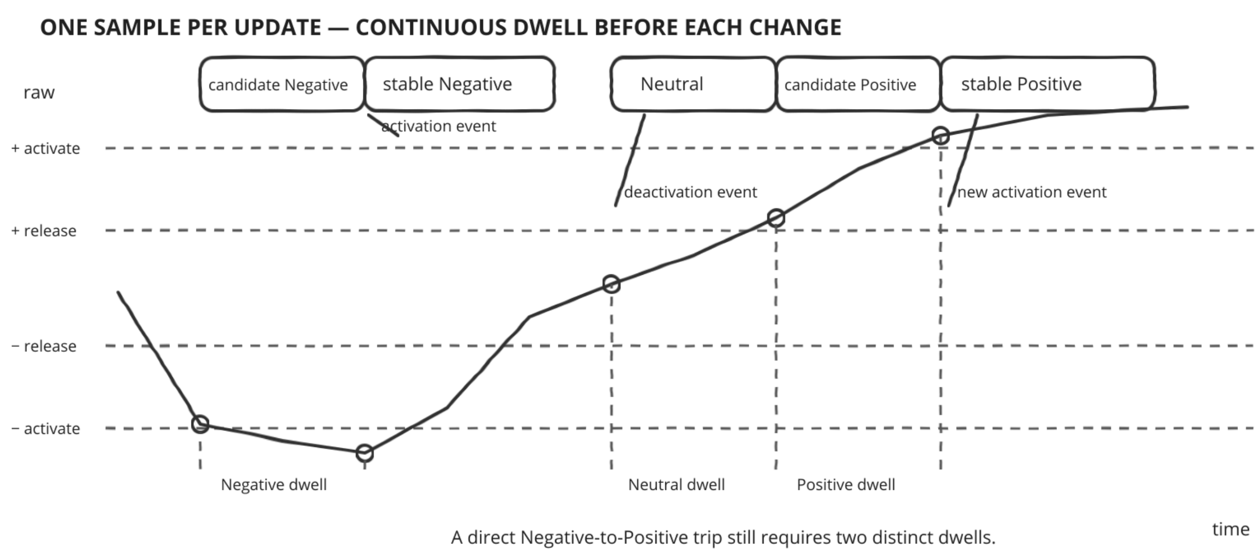
$$q_{\min} \le n_a < n_r < p_r < p_a \le q_{\max} \le 1023.$$

Here n_a, n_r are negative activate/release thresholds and p_r, p_a are positive release/activate thresholds. The separated bands prevent one noisy count from repeatedly changing the stable interpretation.

A candidate polarity must remain continuously present for `dwell`. Travel directly from Negative evidence to Positive evidence first qualifies a Neutral interval and emits deactivation; only a later distinct dwell can emit the new activation. Optional reversal swaps the reported Negative and Positive labels after interpretation. It does not alter the retained raw count.

Snapshot field	Meaning
`raw`, `observedAt`	exact retained ADC sample and caller time
`polarity`	stable Negative, Neutral, or Positive interpretation
`activationEvent`, `deactivationEvent`	transition from the latest update only
`stableFor`	modular duration of the current stable interpretation
`quality`	Valid, below range, above range, or unqualified
`status`	lifecycle or time-contract result

Below- or above-range evidence clears a pending candidate. It does not claim open circuit, short circuit, missing magnet, saturation, or a damaged module. Those hypotheses require independent electrical evidence.



Alternative text: pencil timing plate showing raw samples qualifying Negative after continuous dwell, returning through a separately qualified Neutral dwell with deactivation, then qualifying Positive in a new dwell before activation.

Use the plate as three dependent visible stages: first retain the raw trace, then mark a candidate only while every sample remains in its band, and only then compare the stable observation and event witness. A band exit erases the unfinished candidate; it does not erase the retained sample.

4 Contact observation contract

`MagneticContact` owns one `DigitalInput` with explicit pull, closed level, and dwell. Its canonical raw value is zero for Low and one for High. Its polarity is always `Unspecified`: one contact cannot establish field direction. Constant open or closed evidence likewise cannot distinguish a steady stimulus from a stuck, shorted, or disconnected circuit.

Both components construct inertly, validate before claiming, sample exactly once per explicit-time update, and retain a stable snapshot until the next update. Failed initialization leaks no claim. Repeated initialization and shutdown are idempotent; destruction releases the owned input.

5 Time and deterministic replay

Unsigned modular elapsed time strictly below 2^{31} milliseconds advances. Natural 32-bit rollover is therefore valid. An exact half-range or larger apparent jump is rejected as `InvalidArgument` without sampling or changing transition state. A replay using the same configuration, raw samples, and timestamps produces the same snapshots and pin-operation trace.

6 Narrative Mega example

```
void loop ()
{
    const adk::TimePoint now (millis ());

    observeMagneticInputs (now);

    if (!decideEvidence () || !actuateEvidence ())
    {
        stopSafely ();
    }
}
```

The canonical sketch acquires the two observers and three `MonoLed` witnesses, configures a bounded run, starts with ready evidence, then preserves observe–decide–actuate order. It stops independently after 120 seconds. Serial is unnecessary.

7 Predict, observe, interpret

1. Before connecting anything, predict which marking or source will establish each specimen role. If any role remains inferred, stop.
2. With a qualified, unpowered fixture, predict TP-H and TP-R idle values and each LED's current bound. Check every 1 kΩ resistor.
3. Apply USB power. Observe the ready witness separately from the two raw test points. Interpret ready only as resource-acquisition evidence.
4. Move only the separately authorized retained stimulus through the documented fixture. Predict raw range, dwell, and event order before observing TP-H and the range witness.
5. Exercise the qualified contact without flexing a glass capsule. Compare raw level, stable state, and contact witness through bounce.
6. Invoke shutdown. Predict dark LEDs and released input pins. Measure the named safe state separately; darkness alone does not prove release.

8 Diagnosis and stop conditions

Observation	Stop, inspect, and interpret
unknown marking or pin order	remain unpowered; obtain exact primary evidence
TP-H outside the qualified rail	remove USB; inspect supply and output topology
range witness chatters	compare raw counts, thresholds, dwell, and ground
contact never changes	retain raw level; do not guess stuck versus absent stimulus
event skips Neutral	inspect the trace and separate the two dwell intervals
hot part, odor, reset, or unstable rail	remove USB immediately
loose stimulus or cracked capsule	quarantine the fixture; do not continue

9 Exercises

1. Draw the five ordered thresholds and mark the two hysteresis bands.
2. Explain why reversal changes polarity labels but not raw evidence.
3. Give a rollover trace that advances five milliseconds across zero.
4. Show the reverse rollback order after the second observer cannot claim its pin.
5. List three hypotheses for a constant contact and the additional evidence needed to distinguish them.

Stop rule. Remove USB for heat, odor, reset or rail instability, current-limit trip, out-of-rail output, unexpected movement, specimen/source disagreement, loose stimulus, cracked capsule, or current above a pin, port, package, rail, or aggregate budget. Change wiring only while unpowered. Never create a live short.

10 Software evidence and reference trail

Repository automation verifies threshold boundaries, dwell, hysteresis, reversal, chatter, rollover, backward time, invalid configuration, busy pins, rollback, reuse, ownership traits, Mega compilation, size, and publication. These are host claims, not physical observations.

The searchable HTML lesson is the primary accessible route. Consult the magnetic-observation interface, focused tests, canonical sketch, Lessons 034–036 design record, repository policies, authorized family manifest, and exact-specimen primary sources. This PDF remains untagged and makes no PDF/UA claim.